Coefficient of Friction (Harder Problems)

Exam code: 9MA0

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Coefficient of friction (equilibrium)

Help me understand

How do I model friction when a body is in equilibrium?

- When a body is in static equilibrium under the action of a number of forces, including friction, the value of friction will be less than or equal to the value of μR
- When the value of friction is equal to μR, the body is in **limiting equilibrium** and is on the point of moving
- When a body is stationary on an inclined plane then it is either in equilibrium or in limiting equilibrium, we cannot assume that $F = \mu R$

How do I model friction when a body is moving?

- When a body is moving under the action of a number of forces, including friction, the value of friction will be equal to the value of μR

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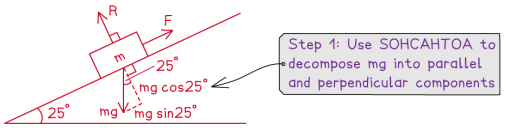
Got it

25° above the horizontal.
The coefficient of friction between the block and the plane is μ. Find the least possible value of μ.

Answer:

Start by drawing a diagram:



Step 2: Find the normal reaction force:

$R = mg\cos 25°$ We don't have a vale for m, so it is easiest to keep R in this exact form

The block is in equilibrium, so the resultant of the forces acting parallel to the plane is zero and $F \leqslant F_{max}$

Step 3: Find F_{max}

$F_{max} = \mu R = \mu(mg\cos 25)$

Step 4: The block is stationary so we do not need to use F = ma. The forces acting parallel to the plane will be equal.

R (↗) $F = mg\sin 25°$

$F_{max} \geqslant F$: $\mu(mg\cos 25) \geqslant mg\sin 25$ (the block will slide down the plane at the point when $\mu R < mg\sin 25$)

$$\mu \geqslant \frac{mg\sin 25}{mg\cos 25}$$

$$\mu \geqslant \tan 25$$

The least possible value of μ is 0.466 (3 sf)

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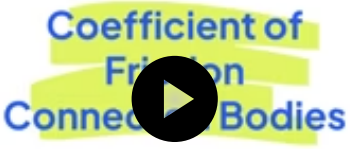
Examiner Tips and Tricks

- Always draw a force diagram and label it clearly.
- Look out for key vocabulary such as rough, smooth, equilibrium, on the point of moving and limiting equilibrium and know how to use each.



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Coefficient of friction (connected particles)

- When two particles are connected by means of a light, inextensible string, friction will act only on any particle that is resting on a rough surface
- On an inclined slope, friction will act in the direction opposite to the direction the particle is moving in or on the point of moving in
- The direction they will move in depends on both the weight of the particles and the angles of the slopes they are resting on
    - Compare the components of the weight in the direction of the motion the particles will move in
    - The particle with the heavier component will move down the slope

Worked Example

Two particles A and B, of identical mass, are connected by means of a light inextensible string. Particle A is held motionless on a rough fixed plane inclined at 30° to the horizontal.

Particle B is resting on a rough fixed plane inclined at 60° to the horizontal. The string passes over a smooth light pulley fixed at the top of the two planes as shown in the diagram below.

Answer:

Given that the coefficient of friction between particle A and the plane is 0.2 and that the particles are in limiting equilibrium, show that particle A is on the point of moving down the slope and find the value of the coefficient of friction between particle B and the plane.

Examiner Tips and Tricks

- Be careful not to assume that the system will move in the direction of the heavier particle, remember to consider the angle too. Make sure your diagram is clear enough to work with, sometimes there can be a lot of information on the diagram.

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